

Chat with DB — Enterprise AI-Powered Analytics Assistant

Technical Design Document (TDD)

1. Project Title

Chat with DB

AI-Powered Enterprise Analytics Assistant for Natural Language Data Querying

2. Project Overview

Chat with DB is an enterprise-grade AI-powered analytics platform that enables business users to retrieve insights from enterprise databases using plain English instead of SQL.

The platform integrates with Amazon Redshift and leverages an internal AI-based Text-to-SQL engine to convert natural language queries into validated SQL queries. The application is designed specifically for operational analytics use cases in pharmaceutical and healthcare enterprise environments.

The primary goal of the system is to reduce dependency on data engineering and analytics teams by enabling self-service data access for business users.

The solution provides:

- Natural language querying
- AI-generated SQL execution
- Controlled and validated query execution
- Human-readable analytics responses
- Audit logging
- Role-based access control
- Query history and saved prompts
- Basic chart and visualization capabilities
- Enterprise-grade security architecture

The MVP version of the platform will initially support:

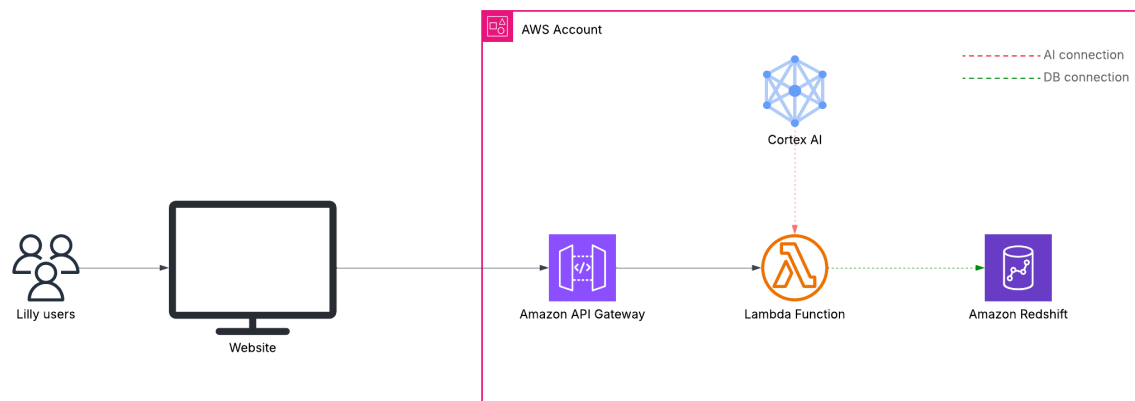
- 3 Redshift tables
- Approximately 1000 rows per table

- Read-only access
- Internal enterprise users only

3. Real-Time Use Cases

#	Use Case	Description
1	Sales Analytics	Business users ask sales-related questions without SQL knowledge
2	Pharma Product Performance	Retrieve product-level operational insights
3	Country-Based Reporting	Analyze metrics by country or region
4	Campaign Monitoring	Monitor AEM tactic and campaign effectiveness
5	Expiry Tracking	Identify records nearing expiry dates
6	Compliance Reporting	Generate compliance-related operational reports
7	Product Adoption Analysis	Analyze product usage patterns
8	Vault-Level Insights	Retrieve vault-specific operational metrics
9	Daily Operational Dashboard	Instant KPI generation
10	Medical Analytics	Understand disease and therapeutic-area trends
11	Approval Workflow Monitoring	Track approved jobs and workflow statuses
12	Self-Service Reporting	Reduce dependency on analytics teams
13	Executive Decision Support	Leadership gets immediate business insights
14	Query Reusability	Save and reuse frequently asked business queries
15	AI-Based Data Exploration	Explore datasets conversationally
16	Trend Analysis	Compare operational metrics across time periods
17	Regional Product Monitoring	Monitor country-level product performance
18	Audit and Governance	Track all executed queries and user activity
19	Ad-Hoc Data Investigation	Instantly answer business questions
20	Business Terminology Mapping	Convert business language into technical database terms

Chat with DB



4. Business Impact and Benefits

Operational Benefits

- Reduces dependency on technical teams
- Accelerates business decision-making
- Enables real-time self-service analytics
- Reduces report generation delays
- Improves operational efficiency
- Increases data accessibility across departments

Financial Benefits

- Reduces manual reporting effort
- Lowers operational overhead
- Reduces repetitive analytics tasks
- Optimizes analyst productivity

Technical Benefits

- Centralized secure analytics access
- Controlled query execution
- Reduced risk of incorrect SQL execution
- Standardized analytics workflows

Strategic Benefits

- Encourages data-driven culture
- Enables faster business insights
- Supports enterprise AI adoption
- Creates foundation for future AI analytics expansion

5. Why This Project is Useful in Real-World Scenarios

Traditional enterprise analytics workflows often require business users to depend on:

- Data engineers
- BI developers
- SQL experts
- Analytics teams

This dependency creates delays in decision-making.

Chat with DB removes this dependency by enabling users to interact with enterprise data conversationally.

Example:

Instead of writing:

```
SELECT country, COUNT(*) FROM sales GROUP BY country;
```

A business user can simply ask:

"Show me product counts by country."

The AI system converts the request into SQL, validates it securely, executes it against Redshift, and returns a readable answer.

This significantly improves productivity and democratizes access to enterprise data.

6. Executive Summary

Chat with DB is an enterprise AI analytics solution that bridges the gap between business users and enterprise databases.

The system uses:

- Natural language processing
- AI-generated SQL
- Controlled query validation
- Secure Redshift integration

The architecture is designed for:

- Enterprise security
- Scalability
- Auditability
- Operational reliability
- Future AI enhancements

The solution is intended primarily for pharmaceutical operational analytics and enterprise reporting.

7. Problem Statement

Business users often face challenges when accessing enterprise data because:

- SQL knowledge is required
- Data teams become bottlenecks
- Report generation takes time
- Ad-hoc analytics are difficult
- Business insights are delayed

This causes:

- Slower decision-making
- Reduced operational efficiency
- Increased dependency on technical resources

The organization requires a secure AI-powered platform that enables business users to retrieve analytics using plain English.

8. Objectives

Primary Objectives

- Enable plain English querying
- Eliminate SQL dependency for business users
- Provide secure Redshift access
- Deliver human-readable analytics results

Secondary Objectives

- Maintain enterprise-grade security
 - Enable audit logging
 - Improve reporting speed
 - Provide reusable query history
 - Enable visualization support
-

9. Scope

In Scope

- Natural language to SQL conversion
- Redshift query execution

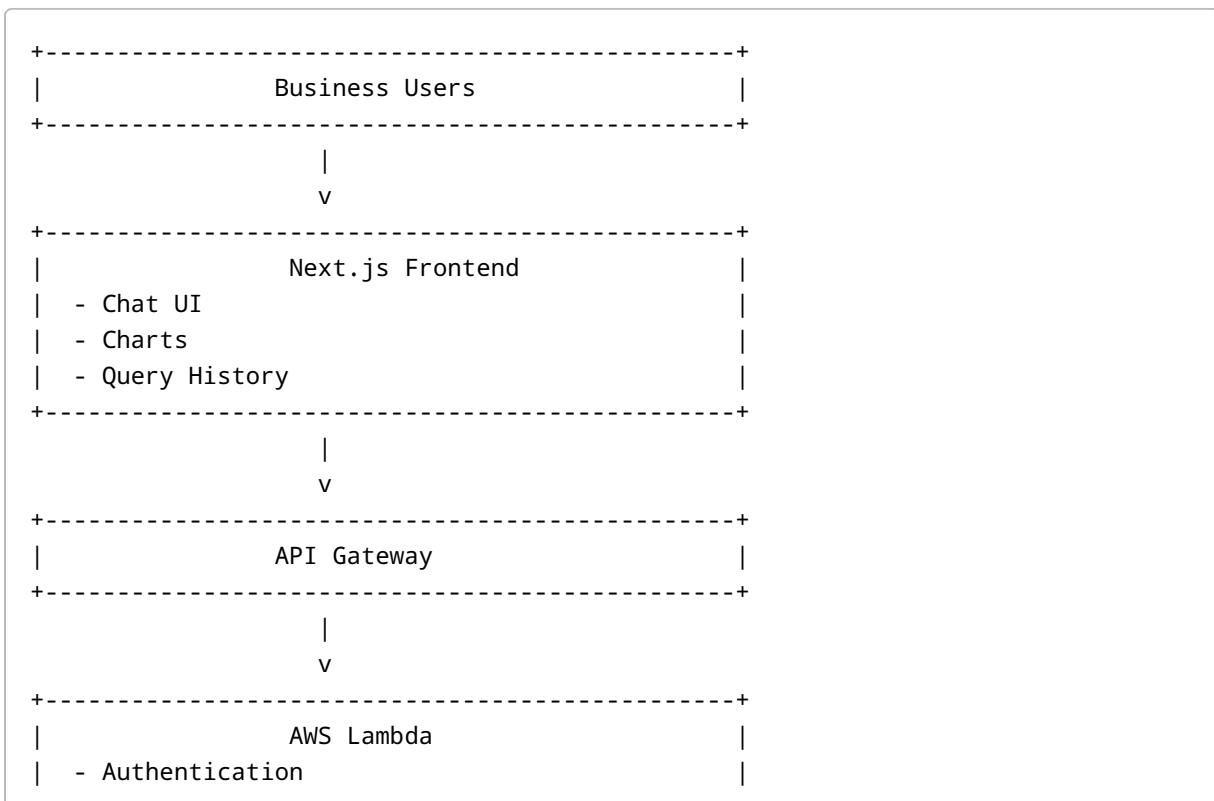
- Read-only data access
- Internal enterprise users
- Query validation
- Query history
- Saved prompts
- Basic visualizations
- Audit logging
- Role-based access

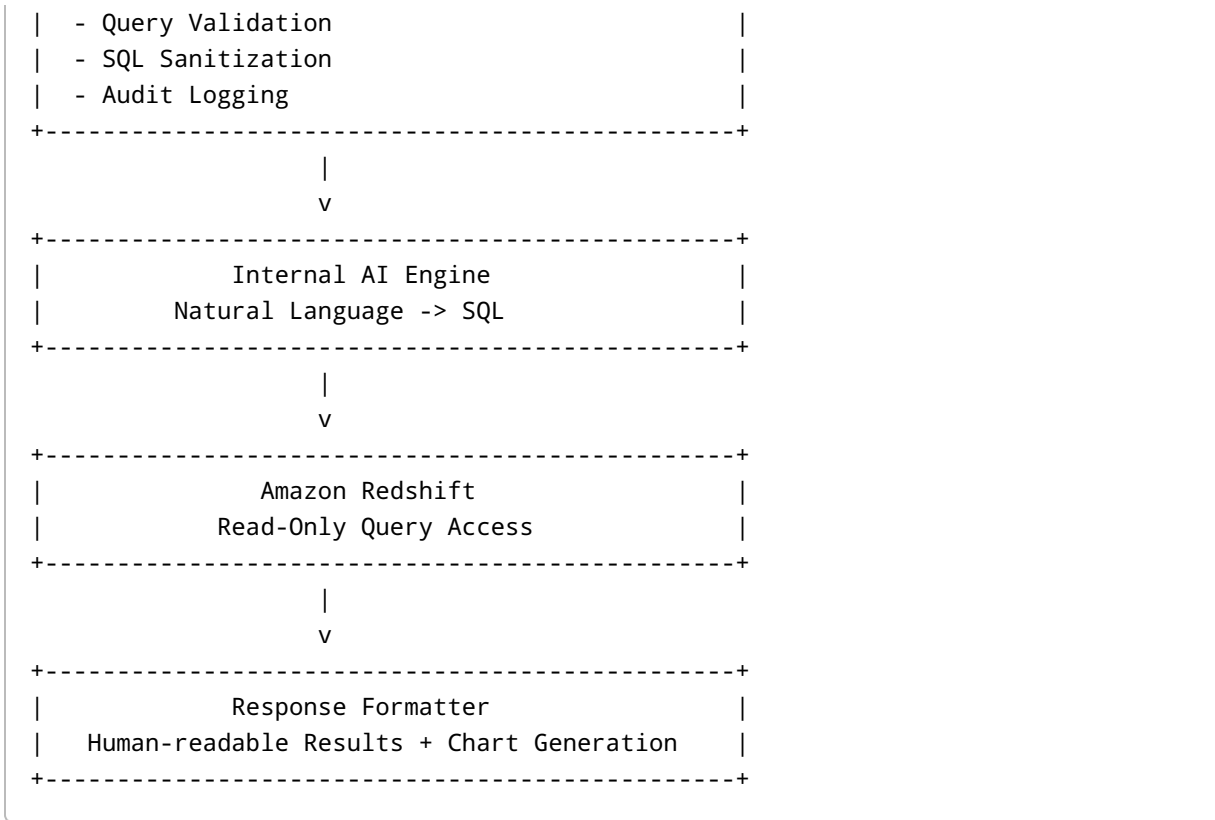
Out of Scope

- Write/update/delete operations
- External customer access
- Multi-database support (MVP)
- Real-time streaming analytics
- Advanced ML predictions
- Voice-based querying
- Mobile application

10. System Architecture

High-Level Architecture





11. High-Level Design (HLD)

Frontend Layer

Responsibilities:

- User authentication
- Chat interface
- Query history display
- Chart rendering
- Saved prompt management

Technologies:

- Next.js
 - Tailwind CSS
 - Shadcn UI
-

Backend Layer

Responsibilities:

- API handling
- Query validation
- AI integration
- Audit logging
- Role validation

Technologies:

- AWS Lambda
 - API Gateway
 - Secrets Manager
 - CloudWatch
-

AI Layer

Responsibilities:

- Convert natural language to SQL
 - Understand business terminology
 - Map business terms to schema fields
 - Generate optimized SQL queries
-

Data Layer

Responsibilities:

- Secure data retrieval
- Read-only access
- Query optimization
- Controlled execution

Technology:

- Amazon Redshift
-

12. Low-Level Design (LLD)

Frontend Components

Component	Responsibility
Chat Window	User query interaction
Query History Panel	Stores historical prompts
Saved Prompt Module	Reusable queries
Visualization Panel	Charts and graphs
User Session Module	Authentication management

Backend Modules

Module	Responsibility
Auth Module	User validation
Query Parser	Process user request
SQL Validator	Prevent unsafe SQL
AI Connector	Connect with internal AI engine
Audit Logger	Log all user actions
Redshift Connector	Execute validated queries
Formatter Engine	Convert results into readable output

SQL Validation Rules

Allowed:

- SELECT
- GROUP BY
- ORDER BY
- LIMIT
- Aggregations

Blocked:

- DELETE

- UPDATE
- INSERT
- DROP
- ALTER
- TRUNCATE
- EXEC

13. Technology Stack

Layer	Technology
Frontend	Next.js
UI Styling	Tailwind CSS
UI Components	Shadcn UI
Backend	AWS Lambda
API Layer	API Gateway
Database	Amazon Redshift
AI Engine	Internal NLP-to-SQL Engine
Secrets Management	AWS Secrets Manager
Monitoring	CloudWatch
Authentication	Enterprise SSO / JWT
Logging	CloudWatch Logs
Infrastructure	AWS

14. Database Design

Sample MVP Tables

Table 1: product_operations

Column	Type
document_id	VARCHAR
department	VARCHAR

Column	Type
country	VARCHAR
product	VARCHAR
disease	VARCHAR
expiry_date	DATE

Table 2: campaign_metrics

Column	Type
tactic_name	VARCHAR
approved_job_id	VARCHAR
classification	VARCHAR
purpose_code	VARCHAR
product_indicator	VARCHAR

Table 3: vault_metadata

Column	Type
vault	VARCHAR
approved_start_date	DATE
product_apims_id	VARCHAR
type	VARCHAR

Database Access Strategy

- Read-only database user
 - Limited schema access
 - Controlled query execution
 - Query timeout restrictions
 - Row limit enforcement
-

15. API Design

POST /query

Purpose: Submit natural language query

Request:

```
{
  "query": "Show product counts by country"
}
```

Response:

```
{
  "sql": "SELECT country, COUNT(*) FROM product_operations GROUP BY country",
  "result": [
    {
      "country": "US",
      "count": 120
    }
  ],
  "summary": "US has the highest product count."
}
```

GET /history

Purpose: Retrieve user query history

POST /save-prompt

Purpose: Save reusable business query

GET /audit-logs

Purpose: Retrieve audit logs for administrators

16. Workflow Diagrams

Query Processing Workflow

```
graph TD
    A[User Query] --> B[Frontend Chat UI]
    B --> C[API Gateway]
    C --> D[Lambda Function]
    D --> E[+--> Authentication]
    E --> F[+--> Query Validation]
    F --> G[+--> AI SQL Generation]
    G --> H[+--> SQL Sanitization]
    H --> I[Redshift Query Execution]
    I --> J[Result Formatter]
    J --> K[Frontend Response]
```

Authentication Workflow

```
graph TD
    A[User Login] --> B[Enterprise SSO]
    B --> C[JWT Token Generation]
```

17. User Roles & Permissions

Role	Permissions
Business User	Query data, view charts, save prompts
Analyst	Advanced analytics access
Admin	User management, audit log access
System Admin	Infrastructure and deployment management

18. Security Considerations

Authentication Security

- Enterprise SSO integration
 - JWT-based authentication
 - Session expiration
 - Multi-factor authentication support
-

Database Security

- Read-only Redshift access
 - Schema-restricted access
 - SQL sanitization
 - Query timeout enforcement
-

API Security

- API Gateway authorization
 - Rate limiting
 - Request validation
 - Payload filtering
-

Infrastructure Security

- VPC-based deployment
 - IAM least privilege policies
 - Secrets Manager integration
 - Encrypted communication (HTTPS/TLS)
-

19. Performance & Scalability

Performance Strategies

- Query caching
 - Connection pooling
 - Optimized SQL generation
 - Pagination support
 - Async query handling
-

Scalability Strategies

- Serverless Lambda scaling
 - Auto-scaling API Gateway
 - Redshift scaling support
 - Stateless frontend architecture
-

20. Logging & Monitoring

Logging

Log Type	Purpose
User Query Logs	Track user queries
API Logs	Monitor API requests
Error Logs	Capture failures
Audit Logs	Compliance tracking
SQL Execution Logs	Query traceability

Monitoring

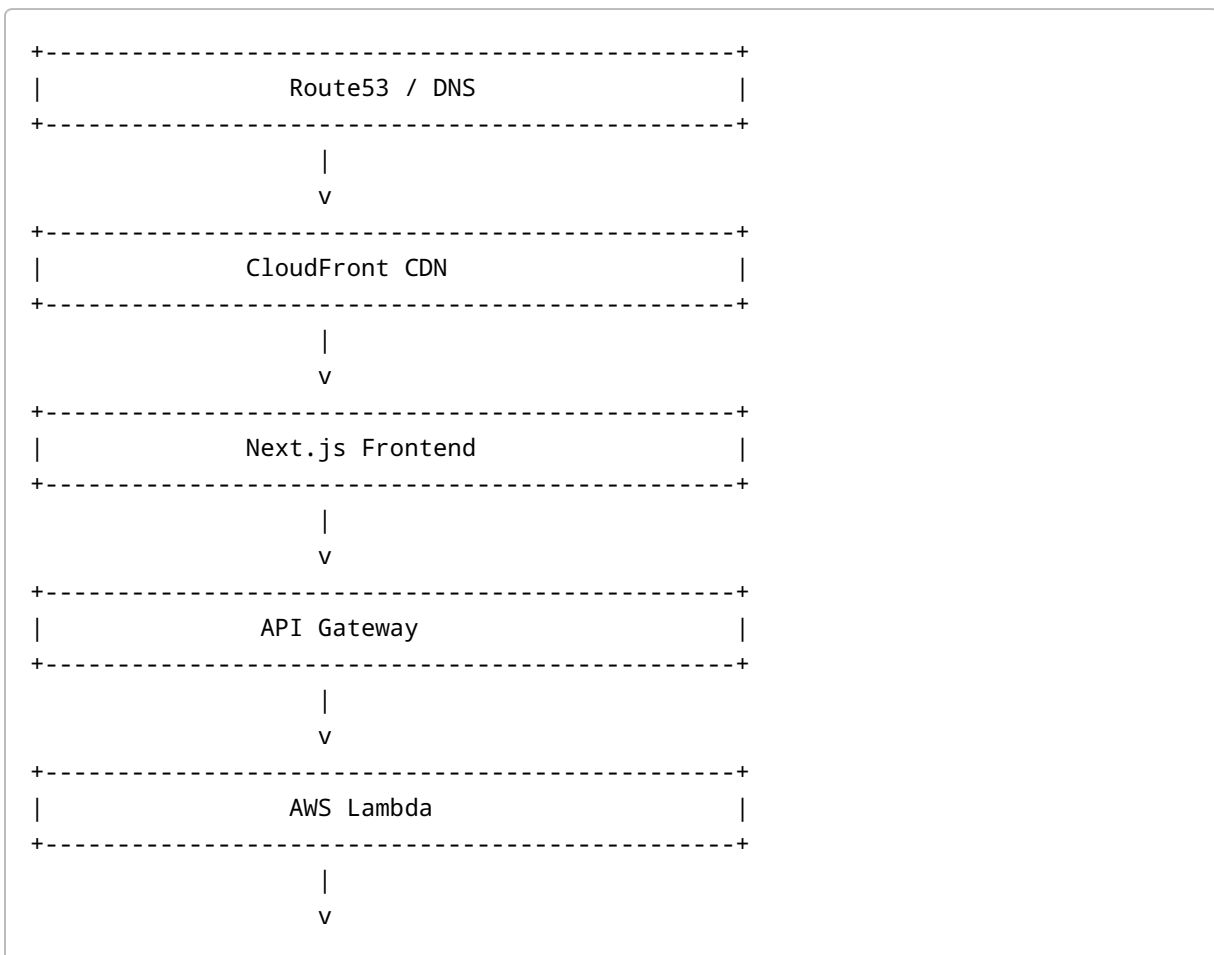
Tools:

- AWS CloudWatch
- CloudWatch Alarms
- Lambda Monitoring
- API Gateway Metrics

Monitored Metrics:

- API latency
- Query execution time
- Error rate
- Failed authentication attempts
- Lambda duration

21. Deployment Architecture



```
+-----+
|               Amazon Redshift               |
+-----+
```

22. CI/CD Flow

CI/CD Pipeline

```
Developer Push
  |
  v
Git Repository
  |
  v
Build Pipeline
  |
  +--> Code Validation
  |
  +--> Unit Testing
  |
  +--> Security Scanning
  |
  +--> Build Artifacts
  |
  v
Deployment Pipeline
  |
  +--> Staging Deployment
  |
  +--> Integration Testing
  |
  +--> Production Deployment
```

Recommended Tools

Purpose	Tool
Source Control	GitHub / GitLab
CI/CD	GitHub Actions / AWS CodePipeline
Infrastructure	Terraform / CloudFormation

Purpose	Tool
Monitoring	CloudWatch

23. Testing Strategy

Unit Testing

- Frontend component testing
 - Lambda function testing
 - SQL validation testing
-

Integration Testing

- API integration testing
 - Redshift connectivity testing
 - AI engine integration testing
-

Security Testing

- SQL injection testing
 - Authentication validation
 - API penetration testing
-

Performance Testing

- Concurrent user testing
 - Query load testing
 - Latency testing
-

User Acceptance Testing (UAT)

- Business workflow validation
 - Query accuracy validation
 - Response readability validation
-

24. Risk Analysis

Risk	Impact	Mitigation
Incorrect AI SQL generation	High	SQL validation layer
Unauthorized data access	High	RBAC and IAM policies
Slow query execution	Medium	Query optimization
AI hallucination	High	Controlled schema mapping
Infrastructure downtime	Medium	AWS high availability
Excessive API usage	Medium	Rate limiting
Sensitive data exposure	High	Read-only controlled access

25. Future Enhancements

Planned Enhancements

- Multi-database support
- Advanced visual analytics
- AI-generated dashboards
- Voice-based querying
- Conversational memory
- Predictive analytics
- Anomaly detection
- Role-based semantic models
- Real-time streaming analytics
- AI-assisted recommendations

26. Recommended Enterprise Enhancements

Semantic Layer

Implement a semantic business layer for:

- Business term mapping
- Synonym support
- Standard KPI definitions
- Controlled metric calculations

Query Governance

Implement:

- Query approval workflows
- Cost-aware query execution
- AI confidence scoring
- Query explainability

Advanced Security

Implement:

- Column-level masking
- Row-level security
- Data classification policies
- Audit retention policies

27. Conclusion

Chat with DB represents a modern enterprise AI analytics platform designed to democratize access to operational data.

By combining:

- AI-driven natural language querying
- Enterprise-grade AWS architecture
- Secure Redshift integration
- Controlled query governance

The platform enables business users to retrieve insights instantly without requiring SQL expertise.

The architecture is scalable, secure, and aligned with enterprise modernization initiatives.

This solution not only improves operational efficiency but also establishes a strong foundation for future AI-driven analytics capabilities within the organization.

28. Appendix

Suggested MVP Milestones

Phase	Deliverable
Phase 1	Basic chat UI
Phase 2	AI SQL generation
Phase 3	Redshift integration
Phase 4	Query validation layer
Phase 5	Charts and visualizations
Phase 6	Audit logging
Phase 7	Enterprise deployment

Estimated Enterprise Benefits

Metric	Expected Improvement
Reporting speed	70% faster
Analytics dependency reduction	60%
Self-service adoption	High
Operational efficiency	Significant improvement
Data accessibility	Enterprise-wide

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